

ITERATORS, GENERATORS, AND OBJECTS Solutions

COMPUTER SCIENCE MENTORS 61A

March 10, 2026 – March 13, 2026

1. What Would Python Display? If you think nothing will be displayed, write N/A.

```
>>> a = [1, 2, 3]
>>> x = iter(a)
```

N/A

```
>>> next(x)
```

1

```
>>> next(x)
```

2

```
>>> y = iter(a)
>>> next(y)
```

1

```
>>> next(x)
```

3

```
>>> next(x)
```

StopIteration Error

```
>>> z = iter(y)
>>> next(z)
```

2

```
>>> [next(y), next(y), next(z)]
```

[2, 3, 3]

2. What Would Python Display? If you think nothing will be displayed, write N/A.

```
>>> x = [1, 2, 3]
>>> y = map(lambda x : x + 10, x)
```

N/A

```
>>> next(y)
```

11

```
>>> [z for z in y]
```

[12, 13]

```
>>> next(y)
```

StopIteration

2 Generators

1. Given the following code block, what is output by the lines that follow?

```
def foo():
    a = 0
    if a == 0:
        print("Hello")
        yield a
        print("World")
```

```
>>> foo()
```

<generator object>

```
>>> foo_gen = foo()
>>> next(foo_gen)
```

Hello
0

```
>>> next(foo_gen)
```

```
World
StopIteration
```

```
>>> for i in foo():
...     print(i)
```

```
Hello
0
World
```

2. Write a generator function that takes in an iterator `it` and yields the running total of the elements produced by it.

```
def accumulate(it):
    '''
    >>> def all_ints():
    ...     i = 0
    ...     while True:
    ...         yield i
    ...         i += 1
    >>> a = accumulate(all_ints())
    >>> [next(a) for x in range(6)]
    [0, 1, 3, 6, 10, 15]
    '''
```

```
def accumulate(it):
    sum = 0
    while True:
        sum += next(it)
        yield sum
```

3. Define `filter_gen`, a generator that takes in iterable `s` and one-argument function `f` and yields every value from `s` for which `f` returns a truthy value.

```
def filter_gen(s, f):
    """
    >>> list(filter_gen([1, 2, 3, 4, 5],
                        lambda x: x % 2 == 0))
    [2, 4]
    >>> list(filter_gen([1, 2, 3, 4, 5], lambda x: x < 3))
    [1, 2]
    """

    for x in s:
        if f(x):
            yield x
```

3 Objects

1. Given the following code, what will Python output for the following prompts? If you think nothing will be outputted, write N/A.

```
class Car:
    wheels = 4
    def __init__(self, gas):
        self.gas = gas

    def drive(self):
        if self.gas <= 0:
            print("Out of gas")
        else:
            self.gas -= 1

    def fill(self, gas):
        self.gas += gas
```

```
>>> c1 = Car(2)
>>> c1.gas
```

2

```
>>> c1.drive()
```

N/A

```
>>> c1.gas
```

1

```
>>> c1.drive()
>>> c1.drive()
```

Out of gas

```
>>> c1.fill(10)
>>> c1.drive()
>>> c1.gas
```

9

```
>>> c1.wheels
>>> Car.wheels
```

44

4 Optional

1. Find an input to the year function that prints the following output: 2 0 2 6

```
def year(a):
    x = iter(a)
    y = iter(a)
    z = iter(x)
    for i in range(next(x)):
        y = iter(a)
        next(y)
    print(next(x))
    print(next(z))
    print(next(y))
    print(next(z))
```

```
year([_, 2, 0, 6, _])
```

There can be **any** number **in** the first blank, **and** there can be **any** number of values after the 6.